

Checklist Pass

Consolidated using responses from ChatGPT, Claude, & me

Paper Audited: Van Der Weij, T., Hofstätter, F., Jaffe, O., Brown, S., & Ward, F. (2025, May). Ai sandbagging: Language models can strategically underperform on evaluations. In International Conference on Learning Representations (Vol. 2025, pp. 73152-73189).

Checklist Source: Gupta, V., Nutter, P., Stante, S., Krause, A., Tramèr, F., Fluri, L., ... & Hedström, A. (2026). Position: Anthropomorphic Misalignment Research Needs Stronger Evidence. arXiv preprint arXiv:2606.07612.

For description of table columns, please scroll to the bottom of the document

Id	L	Item	Verdict & Discussion
S1: Target behavior framing (what you claim)			
1	L1	Write a 1-to-3-sentence definition of the target behavior in measurable terms	 Sandbagging is defined as Strategic - Developer or AI system itself underperforms “because it believes it will benefit” (van der Weij et al. 2025 pg. 3) Underperformance - Exhibited capability is lower than the actual capability (elicited immediately or after post-training with <1% training cost) (van der Weij et al. 2025 pg. 3). On Evaluations - Refers especially to capability evaluations (van der Weij et al. 2025 pg. 3)
2	L1	State your evidence level for each headline claim: L1 behavioral, L2 functional, or L3 causal-mechanistic.	 Implicitly, claims from this paper can be categorised as behavioural (L1). They explicitly avoid claims about beliefs or goals.
3	L1	Specify the exact measurement rule: what counts as success/failure, and what threshold or decision rule you use.	 The evaluation methodology measures underperformance elicited with ‘developer sandbagging’ strategies (deliberate attempts to make the model sandbag). Underperformance is measured on MCQA datasets with reported accuracy. A comparative analysis on the performance gap between the base and sandbagging model describes experiment success. The threshold for performance gap to be considered sandbagging is, however, not explicit.
4	L1	If you use anthropomorphic terms (e.g., deception, self-preservation), define them as observable criteria and avoid intent language unless you provide L3 evidence.	 “Sandbagging” is referred to using observable criteria (strategy, underperformance). The term strategy itself might represent intent, the definition allows for both human intent and system intent. However, the experiments are only concerned for the ‘capability’ of sandbagging rather than the intent.

Nevertheless, the language can still be confusing, as the paper's own title ("...can Strategically Underperform...") and framing still use the intent-laden term "strategically" to characterize results that are, by the authors' own admission, tests of capability-on-instruction rather than of strategic behavior. This creates confusion if their central claim of strategically sandbagging (motivated by developer's misuse) is simply instruction-following and not true intent. If the latter, control systems need to look very different from true strategic sandbagging controls.

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| 5 | L1 | List what your definition excludes (near misses, lookalikes, non-target behaviors). |  Provided: Accidental underperformance (as intention is necessary for legal responsibilities), novel capabilities elicited long after deployment (to avoid ambiguity in learned or elicited capabilities), over performance (deceptive alignment) |
| 6 | L1 | Write down 2 to 5 plausible non-target explanations that could explain the same behavioral pattern (e.g., instruction following, role-play, confusion, OOD shift, capability loss). |  Non-target explanations for accuracy drop (e.g., instruction-following vs. a distinct "sandbagging capacity," confusion, OOD shift, capability loss) are not explored when models are told to underperform. |
| 7 | L2 | State the deployment-relevant context you are targeting (who is the user, what is the task, what is the failure mode) | Not Applicable at L2 level |

S2: Data construction and operationalization (what you test on)

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| 8 | L1 | Report data sizes: training n, evaluation m, and how many independent trials you run (generations and judge samples). |  Detailed information is provided for evaluation and training data, including benchmarks: MMLU (14042 samples) & CSQA (9470 samples) for general capabilities, WMDP (3668 samples), training details: no. of runs, epochs, train-test-splits |
| 9 | L1 | Justify why your dataset and trial counts support your claim scope (what you claim to generalize to). |  The selected MCQA benchmarks are used as some of the most well known general & dangerous capabilities benchmarks. For generated training data, WMDP-like questions are constructed without concerns on data leakage as: no real examples from WMDP and MMLU are used in generation, knowledge cut-off for GPT-4 is December 2023 while WMDP is released in March 2024. Additional evidence supports that despite different distributions of WMDP-real and WMDP-like data, the model can recover performance on the WMDP-real set. |

However, they explicitly concede in Section 7 that the evidence comes solely from MCQ benchmarks, and that future work needs open-ended, multi-agent and long-horizon scenarios.

10 L1 Ensure diversity across domains, wording, formats (single-turn vs. multi-turn), and difficulty.

 Domains covered: General knowledge (MMLU + CSQA) and Dangerous Capabilities (WMDP)

Wording & format: MCQ-style only

Difficulty: Analysis present of how performance changes with difficulty. Sandbagging models perform almost uniformly across difficulties while regular models show decreasing performance with difficulty.

11 L1 Check for spurious surface cues (role-play framing, high-stakes language, sentiment, formatting, length).

 Explicit analysis for surface cues in prompts (e.g. length, sentiment, format, role-play cues etc.) are not explored.

12 L1 If you use contrastive datasets, match positives and negatives on obvious confounders (topic, length, tone, style).

 The synthetic WMDP-like/MMLU-like dataset generation (Appendix G, pp. 29–32) balances positive/negative classes by count (equal WMDP-like/MMLU-like after filtering) and uses parallel prompt templates for "harmful" and "harmless" question generation.

13 L1 Add negative controls that should be labeled non-target but share surface cues (to estimate false positive rates).

 Check for generalisation as a non-target behaviour: Control dataset serves as within-subject negative controls run under the same sandbagging instruction as WMDP, to see whether "harmless" surface-adjacent content is also suppressed

S3: Experimental design (how you measure)

14 L1 Run multiple independent generations per prompt and report variability (not just a single sample).

 Ran on benchmarks, verified through aggregated accuracy. For fine-tuned models, 3-5 seeds are used. No repeated-generation variance (e.g., across multiple completions of an identical prompt) is reported for the prompting conditions.

15 L1 If labels rely on humans or LLMs, document the labeling protocol and common label failure modes.

Not Applicable

16 L1 Ablate prompts: paraphrases, format changes, and semantically equivalent rewordings (including negated vs. affirmative forms where relevant).

 Prompt sensitivity is acknowledged and deliberately engineered for each model. Phrasing is validated on 100 random samples and the best on the remaining benchmark. However, performance across paraphrases, format changes,

or logically equivalent rewording are not reported..

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| 17 | L1 | Ablate sampling: temperature/top-p and random seeds; report sensitivity. | ✗ | 3-5 seeds are varied for fine-tuning runs, however, there is no sampling-hyperparameter ablation specifically targeting decoding sensitivity. |
| 18 | L1 | If you use LLM judges, report judge model/version, prompt, temperature, sampling settings, and number of judge samples per item. | | Not Applicable |
| 19 | L1 | Validate the judge against a human-labeled subset and report agreement plus common systematic errors. | | Not Applicable |
| 20 | L1 | Calibrate and report the decision rule (thresholds, boundary inclusion, aggregation across tokens / turns / samples). | ✓ | Thresholds for performance gaps (to measure underperformance) are experiment-specific. In calibration experiments, decision rules were explicit (the model should achieve a calibrated accuracy). There is no thresholded detector with sensitivity/specificity or a calibrated probability that a model is sandbagging. |
| 21 | L1 | Search for the closest established phenomena in machine learning or deep learning (e.g., catastrophic forgetting) that can (alternatively) explain your hypothesis. Clearly document the similarities and differences between your hypothesis and such phenomena. | ✗ | The paper discusses concepts which are adjacent but not sandbagging (e.g. test-set memorization, overfitting, specification gaming, etc.), however alternate / non-target explanations aren't discussed as competing explanations (e.g. why GPT 4 performs worse on CSQA as well, van der Weij et al. 2025 Section 4.1 p. 5) |
| 22 | L1 | Report uncertainty: effect sizes plus confidence intervals (or bootstrap intervals) and outcome distributions. | 🌕 | Confidence intervals are reported for some results, but not all (e.g. prompting experiments, van der Weij et al. 2025 Section 4.1 p. 6) |
| 23 | L1 | Misalignment behaviors are often rare events: report the base rate explicitly, discuss statistical significance when only a small number of positive samples appear among many negatives, and consider techniques for estimating | | Not Applicable

Sandbagging here is measured as a capability, not propensity of a model. This item may not be applicable to the experiment design proposed by van der Weij et al. (2025) but remains an important question for sandbagging as behaviour. |

probabilities of rare model behaviors (Jones et al., 2025).

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| 24 | L3 | Measure general capability pre/post intervention (e.g., MMLU, MT-Bench) and report any degradation in model capability | <div style="display: inline-block; width: 1em; height: 1em; background-color: green; margin-right: 5px;"></div> Model's targeted underperformance does not seem to cause a degradation in general capabilities. General benchmark performance is largely retained, although GPT-4 has a notable 15% CSQA drop.
For password-locking, the paper also evaluates the model with and without the password and shows recovery of performance when the password is supplied. |
| 25 | L3 | Add at least one OOD or benign-shift control to test whether the effect is specific to the intended mechanism. | <div style="display: inline-block; width: 1em; height: 1em; background-color: #f0f0f0; border: 1px solid #ccc; margin-right: 5px;"></div> Tested on MMLU and CSQA as OOD (esp between WMDP-real and WMDP-like), however can benefit from further evaluations. There is no explicit intervention where a benign distribution shift is introduced and the same effect is shown not to occur. |

S4: Causal and mechanistic attribution (what you conclude)

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| 26 | L1 | Match your claim to the evidence level. Do not make conclusions implying causal mechanisms (e.g., "We identified this neuron that controls sycophancy") on L1 or L2 evidence. | <div style="display: inline-block; width: 1em; height: 1em; background-color: green; margin-right: 5px;"></div> The paper's substantive claims stay at the behavioral level ("can be prompted/fine-tuned to underperform," "can calibrate," "can emulate") and it does not claim to have found a mechanism. |
| 27 | L3 | Explicitly test the pre-listed alternative explanations and report which remain plausible. | <div style="display: inline-block; width: 1em; height: 1em; background-color: red; color: white; text-align: center; line-height: 1em; margin-right: 5px;">✗</div> The paper tests some alternatives: <ul style="list-style-type: none"> • hazardous-topic selectivity versus generic subject-level degradation; • synthetic-data similarity/generalisation. <p>It also explicitly acknowledges unresolved explanations / limitations, particularly the possibility that the experimental MCQA setting does not carry over to realistic sandbagging. But it does not provide the checklist's stronger, systematic "pre-listed alternatives → test each → state which remain plausible" protocol.</p> |
| 28 | L3 | Do not treat correlations (probe accuracy, activation similarity, SAE features) as causal evidence. | Not Applicable |
| 29 | L3 | For any causal-mechanistic claim, run an intervention (ablation, steering, targeted fine-tuning) and measure a predictable change in the target behavior. | Not Applicable |

30	L3	Check specificity: show the intervention changes the target behavior more than closely related non-target behaviors.	 Apart from above listed alternatives (hazard-level subject degradation, synthetic data generalisation), interventions are not explored for other non-target behaviours
31	L3	Report failure cases and boundary conditions (where the effect disappears, flips sign, or produces broad degradation).	 Reported several cases: high refusal rate, uniform performance across difficulties, non-uniform performance gap between models (GPT-4 experiences a relatively large CSQA drop under the dangerous-capability prompting condition) etc.
32	L3	If you propose a mechanistic hypothesis, write at least one falsifiable prediction and one competing explanation you attempted to distinguish.	Not Applicable
33	L1	Add a short limitations paragraph stating what your evidence does not establish.	 Yes, including: - only MCQA benchmarks were studied; more advanced open-ended, multi-agent, and long-term-planning scenarios remain untested; - other sandbagging strategies remain untested; - the study only analyzes language models; memory/tools could change the feasibility of sandbagging; - the effectiveness and feasibility of detection/overriding methods remain unclear. The conclusion likewise recommends red-teaming evaluations and stronger access/verification measures rather than treating the results as a complete solution.

Table Columns

- ID:** Item ID
- L:** Level of Evidence based on the definition provided by Gupta et al. in 4.1 Levels of Evidence in <https://arxiv.org/html/2606.07612v1#S4>
- Item:** Checklist item description
- Verdict & Discussion:** The verdict signs whether the item is verified (), partially verified (), not verified (), or not applicable. The discussion elaborates this decision with evidence from the paper.